

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Efficient data processing and machine-annotated text, in production and in research

Research interest for this post: indexing and query processing for machine-annotated text: indexes that answer keyword, label and score-range queries efficiently and stay correct when a new model version relabels the corpus (project outline attached). Over seven years: 5+ companies and 300+ freelance clients.

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EDUCATION

2024 - 2025 awarded Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Credit-weighted average 70.5/100 (Distinction threshold 70). Master's project 78/100. Supervisor: Dr Ella Haig. 180 UK credits, conventionally mapped to 90 ECTS. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, Explainable AI, Data Management (80). Taught and assessed in English.
2017 - 2021 awarded Jan 2022	BSc Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan. Four years full time, 130 credit hours completed. CGPA 2.75 on a 0.00 to 4.00 scale. Courses included Database Systems (B+), Data Structures, Design and Analysis of Algorithms, Operating Systems, Compiler Construction, Theory of Automata and Data Mining.

MACHINE LEARNING IN PRODUCTION

Feb 2025 - present	Founder and Chief Technology Officer, ArtemisAI Ltd (trading as ViralData), United Kingdom (remote). Running in production ahead of a December 2026 launch. • Six fine-tuned encoder classifiers for sentiment, emotion, toxicity, topic, language and intent, built from XLM-R, DistilBERT, DeBERTa-v3 and Detoxify bases and served from six SageMaker endpoints, called in parallel under 100ms. Custom per-sample loss weighted by label confidence, class balance and label source. • LLM labelling and distillation chain that produces the training data: Claude labels a stratified seed of about 6,000 posts together with its reasoning, the decision rules in that reasoning are mined into a prompt for DeepSeek, DeepSeek labels about 12,000 more; both tiers were first checked against about 500 hand-labelled posts and comments. Result is about 18,000 machine-labelled items, which with synthetic cases makes a roughly 20,000-item confidence-weighted dataset, stratified 80/20. The models I fine-tune are encoder classifiers; I use Claude and DeepSeek through APIs and have not fine-tuned a generative model. • Deploy gate and retraining loop. A release ships only at macro-F1 above 0.85 on both a 97-post human holdout and a 4,000-item stratified test set, followed by a human check of 200 random posts. Weekly routing loop with three-tier confidence routing: roughly 75% auto-committed above 0.85, 18% sent to review between 0.60 and 0.85, 7% held below 0.60. Weekly KL-divergence drift alarms on output distributions and CI quality gates on every release. • Images and video: PyTorch fusion classifier over CLIP embeddings of image, OCR text and caption (macro-F1 0.93 on a gold set); hybrid OCR pipeline (token-F1 0.928); Whisper transcription; a networkx affinity graph for a collaboration feature. • Data platform underneath the models: Meta Graph API ingestion of 30+ public pages daily into a layered lakehouse and Redshift warehouse, 111,000+ posts and 3.39 million comments processed. Products built on top: underperforming-post prediction, posting-time recommendation, drafting and crisis alerts. • Versioned labels. Reported accuracy moved with the reference (v1 to v3 scored against a 97-post human holdout, v4 onward against Claude's labels), and the two annotation tiers agree on 82.9% of sentiment but 74.4% of emotion labels: an index over such labels must know which model version produced them.
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MACHINE LEARNING IN RESEARCH

Master's thesis: Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies. Fine-tuned a BERT classifier to **macro-F1 0.878** (later found to be scored against the lexicon-seeded labels, not the human set; the manuscript says so) against BiLSTM and lexicon baselines, with human validation at Cohen's $\kappa \approx 0.78$, temperature-scaling calibration of the predicted probabilities, McNemar and bootstrap significance testing of model differences, chi-square topic tests, LIME and integrated gradients for interpretability, NRC emotion analysis linking anger and fear to negative policy reactions, and Prophet forecasting of the daily sentiment index (MAE ≈ 0.02). Collected $\sim 279,000$ public posts on UK economic policy from X/Twitter, Reddit, YouTube and Quora via official APIs and licensed feeds, under ethics approval (Portsmouth TETHIC-2025-111094) and GDPR-compliant procedures. *The finding that shaped my research interest was not the accuracy: the same policy event produced structurally different sentiment and emotion profiles on different platforms, which meant the cross-platform aggregate rested on an untested assumption, that the classifier measures the same construct everywhere it reads.*

RESEARCH OUTPUT

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." **Manuscript, complete; not published and not submitted.** Treats each social platform as a survey mode and applies generalizability theory, measurement invariance and differential item functioning, and score linking to machine-labelled text. In a 500-replication simulation with known ground truth, an index that pools platform-level scores at face value reports the wrong sign of the aggregate on roughly a quarter of days, purely from measurement differences between platforms.

DATA ENGINEERING AND SOFTWARE

Dec 2024 - present	Data Engineer, consultant, M+C Saatchi Fluency, United Kingdom (remote) - Analysing online data for brands and supporting UK Government social listening pipelines. Built the Amazon Sprinklr ETL (cleaning, deduplication, translation, tagging into Postgres and Tableau), with a composite key so that a 770,000-row weekly pipeline re-translates only unseen records. Clients include Amazon, Ford, Nike, Reckitt, Abbott and CFMOTO; AWS (ECS Fargate, Step Functions, Glue, Athena). 68 collectors ingesting ten platforms daily into a governed 156-column master table; 6.7 million items in a six-week window at 99%+ uptime. Automated repair agent as a named CI workflow: when a source changes shape the system diagnoses the break, regenerates the extraction logic and tests it against live pages before shipping. - Anti-bot handling, proxy rotation and TLS fingerprint management; weekly KL-divergence drift alarms on pipeline output distributions; documented filter rules, removal lists and pre-delete backups on government-facing datasets.
Feb 2022 - Aug 2024	Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan Led a three-engineer team owning 500+ regional collectors across 15 MENA countries on Kubernetes/EKS, gathering multilingual platform data in English, Arabic, Urdu and Hindi. Redesigned the layered warehouse (RDL, ODL, ADL), improving ETL/ELT throughput by 60%. Built competitor-analysis dashboards from e-commerce and real estate listings and online discussions that predicted revenue and showed what customers said. Promoted twice in two years.
2018 - present	Earlier and concurrent engagements Fix.com, Canada, consultant, May 2023 to present (part time): Scrapy/Scrapyd pipelines processing 50M+ data points from 200+ sources; Great Expectations validation with automated schema-drift detection, cutting bad production data by 70%. VendueTech, Ireland, Lead Data Engineer, Jun to Jul 2025: led three engineers on real-time auction ingestion, migrated Azure to AWS. Prefe, Italy, consultant, Sep to Dec 2024: rebuilt legacy collectors, cutting failure rate 60%. CXG, Dubai, Dec 2025 to Jan 2026: production collectors for luxury retail at 99%+ completeness against advanced anti-bot protections. Freelance data engineer (Level 2 seller), Dec 2018 to Feb 2026: 300+ five-star projects across 400+ websites for 300+ international clients.

METHODS AND TOOLS

Machine learning	Transformer fine-tuning (BERT, XLM-R, DistilBERT, DeBERTa-v3, Detoxify); LLM-assisted labelling and distillation into smaller encoder models; confidence-weighted loss and class-imbalance handling; CLIP multimodal embeddings and fusion networks; Whisper transcription; OCR pipelines; BiLSTM; PyTorch; SageMaker training, endpoints and weekly retraining loops.
Evaluation	Macro-F1 against human holdouts and stratified test sets; inter-coder reliability (Cohen's κ); temperature-scaling calibration; McNemar and bootstrap testing; chi-square; KL-divergence drift detection; LIME and integrated gradients; Prophet forecasting.
Measurement	Generalizability theory, measurement invariance and differential item functioning, score linking; treating annotators and platforms as measurement instruments rather than as ground truth.
Data collection	Scrapy, Playwright, Selenium; official platform APIs (Meta Graph, Reddit, YouTube); licensed feeds; proxy and anti-blocking management; TLS fingerprinting; detection anatomy; agentic browsers.
Engineering	Python, PySpark, SQL; AWS (SageMaker, Fargate, Glue, Athena, Redshift, Step Functions, Lambda, DynamoDB); Airflow; Docker and Kubernetes; PostgreSQL; CI/CD with automated quality gates. R functional rather than fluent. Production code is Python; building C++ systems skills (on the PISA engine) is the first step of my PhD plan.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25, the UK Foreign, Commonwealth and Development Office's flagship global scholarship, selected from over 70,000 applicants across 160 countries and awarded to fund the MSc at Portsmouth.
Public work	Author and maintainer of web-scraping-guide.com , a free public engineering reference on web data collection: 24 sections covering 60+ libraries and tools, six anti-bot systems, detection anatomy, agentic browsers, legal and ethical constraints, production architecture and data quality.
Languages	Urdu (native). English: MSc taught and assessed in English, IELTS Academic 6.5 overall, March 2024. No Norwegian yet; I would begin learning on arrival (NTNU offers A2 training).
Teaching	No formal university teaching. Mentored two junior engineers at Prefe and led three-engineer teams at Dubizzle Labs and VendueTech; wrote and maintain a public technical reference used by practitioners.